



NITI Aayog



SECTOR AT A GLANCE 15

Integrated Energy Assessment in India

Energy in India

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INTEGRATED ENERGY ASSESSMENT IN INDIA

ENERGY IN INDIA

The '**Sector-at-a-Glance**' provides an easy-to-read overview and basic understanding of a sector, sub-sector, or thematic area. The facts and insights on the sector are supported by relevant data and best practices from credible government and multilateral sources. The document covers the structure, key trends, opportunities and challenges, government initiatives, and the way forward for the sector.

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Overview

Integrated energy assessment provides data and insights to guide decisions related to energy sources, production and trading, and environmental regulations. Additionally, and crucially it also provides policy relevant insights into sustainable development and environmental change issues by complementing socioeconomic modelling. This requires modelling and evaluating the energy system in coordination with multiple related systemic factors such as economic growth, resource availability, air pollution, and land-use analysis. It facilitates holistic planning, target setting for the short and medium term, and policy design to strengthen action at central and state levels across various ministries, departments, or agencies. Integrated energy assessment also evaluates and identifies the preferred technology and policy options that help achieve the objectives of energy security, improved access, affordability, efficiency, resource, and environmental sustainability. In India, the Prime Minister's Office and the NITI Aayog, in collaboration with various central and state ministries and departments, work collaboratively to set a roadmap towards achieving these shared goals.

Adopting an effective integrated energy assessment can ensure that India achieves universal access to affordable electricity and clean cooking, ensuring energy security and transitioning to renewable energy within a specific timeframe and in a cost-efficient and sustainable manner.

However, addressing challenges such as limited data availability for non-commercial energy sources like biomass, lack of coordination between institutions in India's energy sector, and the need for capacity building of policymakers is critical to conducting holistic energy planning and policy design.

Given these opportunities and challenges, integrated real-time energy data, institutionalised inter-ministerial collaboration, and a dedicated integrated energy policy and its successful implementation are crucial to ensure effective integrated energy assessment in India. Several ongoing and recent central government initiatives, including but not limited to the Integrated Energy Policy, and Energy Conservation Act focus on establishing country-wide uniform guidelines for energy security and energy efficiency. These efforts, along with several state- and local-level policies and initiatives, seek to enable India to achieve its targets for energy transition, access, affordability, and security in a timely and cost-effective manner.

Enhancing energy policy coordination across central and state ministries, advancing renewable energy and energy access planning, and investing in capacity-building and data systems for improved energy governance will be critical to continue advancing India's energy ambitions. This note presents key trends related to different components of integrated energy assessment in India and actions for the way forward.

Figure 1: Key integrated energy assessment statistics

Net zero

Emissions targeted by 2070 by GoI

2.5-3B tonnes

Of CO₂ equivalent additional carbon sink creation targeted by 2030 by GoI

45%

Reduction in emissions intensity of India's GDP targeted by 2030

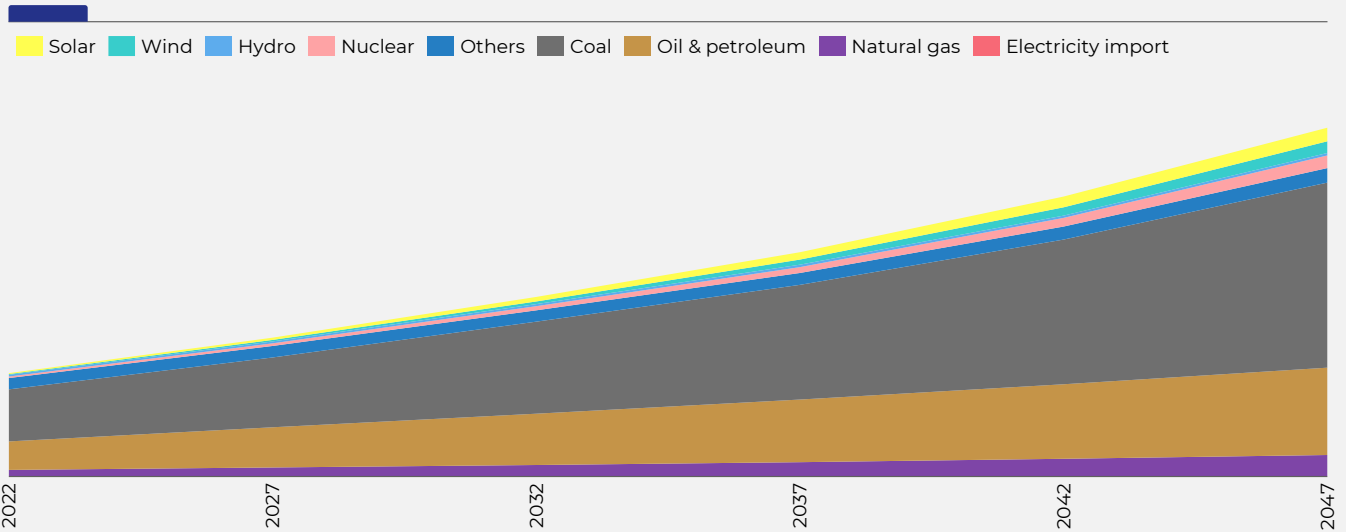
50%

Of total electric power installed capacity fulfilled from non-fossil fuel sources target by 2030 by GoI

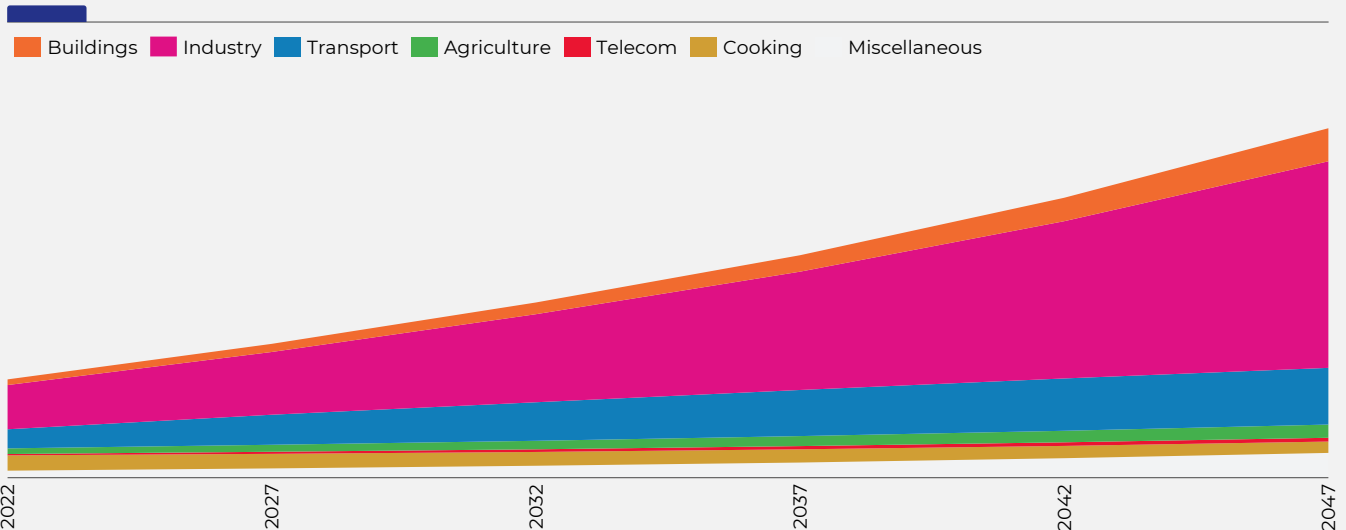
Source: Ministry of Environment, Forest and Climate Change, India's Long-Term Low-Carbon Development Strategy, 2022; Cabinet, Cabinet approves India's Updated Nationally Determined Contribution to be communicated to the United Nations Framework Convention on Climate Change, 2022

Figure 2: India's energy demand and energy supply, 2022-2047

India's energy demand between 2022 and 2047



India's energy supply between 2022 and 2047



Source NITI Aayog, India Energy Security Scenarios 2047, accessed January 2024

Opportunities and challenges

Opportunities

This section highlights the outcomes policymakers can achieve through effective integrated energy assessment. These revolve primarily around ensuring smooth energy transition, ensuring long-term energy security and ensuring universal and affordable access.



Improving decision making and resource management for a sustainable and smooth energy transition:

By conducting a comprehensive evaluation of the country's energy landscape, integrated energy planning can identify the most viable pathways for transitioning from traditional fossil fuels to sustainable and renewable energy sources while managing interconnected challenges like the water-food-energy nexus and the climate-land-energy-water interconnections. Such assessment can play a crucial role in addressing climate adaptation goals, by providing in-depth analysis of local conditions, barriers, and enablers for a clean energy transition. It also helps in setting clear and realistic state-specific energy transition goals, developing appropriate policies, and aligning the efforts of various stakeholders towards common targets. It also addresses uncertainties related to resource availability and energy prices and offer valuable insights for short, medium, and long-term policy formulation, allowing governments and other relevant stakeholders to effect structural shifts that boost economic growth and other development objectives while ensuring a smooth energy transition. For example, as the sixth and most recent Intergovernmental Panel on Climate Change report highlights, integrated approaches and cross-institution coordination – such as a water-energy-food nexus approach – can help policymakers evaluate competing policy objectives, reduce trade-offs, and allow for greater synergies across sectors.¹



Ensuring energy security:

The NITI Aayog in close coordination with ministries and departments at the central and state level can help evaluate India's energy demand and supply and identify potential vulnerabilities and risks to the energy sector. This assessment enables policymakers to develop robust strategies for diversifying energy sources, enhancing energy efficiency, and reducing dependence on imported fuels. They can also facilitate long-term planning and investment decisions, promoting the development of indigenous energy resources and strengthening energy infrastructure. As an example, the NITI Aayog created the India Energy Security Scenarios 2047 (IESS 2047) calculator and the India Climate and Energy Dashboard (ICED) to involve diverse stakeholders in energy planning and promote well-informed decision-making at various levels.² These tools aid consistent and holistic decision-making.



Ensuring universal and affordable energy access:

By conducting a thorough assessment of the nation's energy requirements and available resources, integrated assessments can assist in identifying regions with insufficient access to energy services and varied availability of resources. This evaluation facilitates the formulation of focused and inclusive policies to address the energy disparity in underserved

¹ Caretta, M.A., A. Mukherji, M. Arfanuzzaman, R.A. Betts, A. Gelfan, Y. Hirabayashi, T.K. Lissner, J. Liu, E. Lopez Gunn, R. Morgan, S. Mwanga, and S. Supratid, 2022: Water. In: Climate Change 2022: Impacts, Adaptation and Vulnerability. Contribution of Working Group II to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change; IPCC, Mitigation of Climate Change: Working Group III contribution to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change, 2022
² Lok Sabha Secretariat, Energy Security, 2017; NITI Aayog, India Energy Security Scenarios 2047, Accessed September 2023; NITI Aayog, India Climate and Energy Dashboard, Accessed September 2023

areas and marginalised communities. Moreover, integrated assessments help identify suitable technologies and solutions that cater to diverse energy needs, including renewable energy sources and off-grid alternatives for remote locations, thereby playing a pivotal role in achieving universal energy access.

Challenges

This section highlights some of the major challenges and constraints related to



Limited state-level non-commercial energy data collection, sharing, and management:

While platforms like ICED and IESS provide a comprehensive national-level view of India's energy scenario, a state-level picture is absent. Further, as per the Bureau of Energy Efficiency's (BEE) recent report on National Energy Data, there is limited data available for non-commercial energy sources like biomass, despite their substantial contribution to energy requirements.³ Moreover, while historical data is publicly accessible, there are multiple sources for a single variable, each providing different data points. For instance, various sources report state-wise sectoral electricity consumption, such as the state Economic Survey, the CEA Dashboard, and the Report on Performance of Power Utilities by Power Finance Corporation (PFC) Ltd. The complexity arises from the fact that these agencies often employ different methodologies, resulting in varying estimations.⁴



Limited coordination between institutions in India's energy sector:

Integrated energy assessment entails coordinated efforts from multiple stakeholders and decision-makers both within and outside the energy sector, and spanning policymakers, industry actors, and civil society. This is essential to address potentially competing objectives while also adapting to a dynamic environment where economic conditions and regulatory responses are in flux. As the energy sector undergoes significant transformations, related sectors and industries of the economy must also adapt their operations in step. Such entities often have distinct sets of targets and objectives, making it essential to ensure alignment towards a common goal. Achieving this alignment requires a combination of mandates, incentives, and penalties, supported by robust accountability and monitoring mechanisms. While progress in this regard is clear, achieving this cohesive coordination remains an ongoing challenge, as noted by the International Energy Agency.⁵ Enhancing working-level coordination among various ministries across sectors, and institutionalising current coordination mechanisms also pose significant challenges. Reconciling the diverse interests and perspectives of different ministries in a timely manner requires careful attention. Establishing an institutionalised coordination mechanism becomes crucial to overcoming these challenges, ensuring efficient and prompt resolution of issues in the energy sector.⁶



Lack of dedicated capacity building for policymakers:

As reflected in NITI Aayog's 2022 SECI state scorecard, Indian states can continue to advance well-coordinated energy plans in concert with the central government to ensure energy access, affordability, efficiency, reliability and sustainability for all citizens.⁷ This is critical for the unfolding energy transition, which requires that states undertake several well-harmonized efforts to advance clean energy.⁸ However, currently states lack inter-departmental coordination and knowledge sharing which is essential given the cross-

³ Bureau of Energy Efficiency, National Energy Data: Survey and Analysis, 2023

⁴ CEEW, How State Climate Plans Are Key to India's Net-zero Target, 2023

⁵ International Energy Agency, Understanding Energy Challenges in India, 2012

⁶ International Energy Agency, Understanding Energy Challenges in India, 2012

⁷ NITI Aayog, State Energy and Climate Index, 2022

⁸ IEEFA, Indian States' Electricity Transition, 2023

sectoral nature of decarbonisation and integrated energy.⁹ Together, these underscore the need for dedicated capacity building and assistance to policymakers including participatory planning and budgeting, community participation, knowledge and expertise sharing among different departments of the states to develop well-coordinated and cohesive integrated energy roadmaps that are in line with national and state targets.¹⁰

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⁹ CEEW, How State Climate Plans Are Key to India's Net-zero Target, accessed on December 2023
¹⁰ NITI Aayog, Participant's manual: Capacity Building Program for Civil Servants to Implement

Institutional framework and policy landscape –

Key ministries and departments

This section highlights the major central government bodies and national-level institutions that oversee integrated energy assessment in India.



NITI Aayog

The NITI Aayog serves as the apex public policy think tank of the Government of India. Within the energy vertical, it deals with five key sectors: power, coal, petroleum and natural gas, new and renewable energy, and atomic power. It conducts policy analysis of these sectors and undertakes steps to promote integrated energy assessments in India. For example, the NITI Aayog created the India Energy Security Scenarios 2047 (IESS 2047) calculator and the India Climate and Energy Dashboard (ICED) to involve diverse stakeholders in energy planning and promote well-informed decision-making at various levels. These tools aid consistent and holistic decision-making.



Ministry of Coal

The ministry determines policies and strategies in respect of exploration and development of coal and lignite reserves, sanctioning of important projects, and conducting assessments to collect and share data.



Ministry of Power

The ministry administers, formulates, and implements power sector-related policies and provides R&D and technical assistance relating to grid integration, transmission system networks and distribution systems in the States/UTs.



Ministry of New and Renewable Energy (MNRE)

The ministry develops renewable and clean energy technologies, systems, products, and services in line with global standards and deploys these for energy applications for sectors like power, transport and support country's objective of achieving energy security.



Bureau of Energy Efficiency

The bureau is the nodal authority for all GH2 projects in India, assists in developing policies and strategies to promote energy efficiency and conservation and utilises existing resources and infrastructure in performing functions under the Energy Conservation Act.



Ministry of Petroleum and Natural Gas

The primary focus of the ministry is on the exploration, production, refining, and marketing of petroleum and natural gas along with energy policy and planning via energy assessments in the country.



Prime Minister's Office (PMO)

The PMO provides secretarial assistance to the Prime Minister of India through strategic direction, coordination, and high-level leadership, critical for shaping energy policies through energy assessments.



Ministry of Environment, Forest and Climate Change (MoEFCC)

The MoEFCC develops and implements policies and programs related to climate change mitigation and adaptation, environmental impact assessments, and sustainable development. This ensures that energy development aligns with national environmental goals and contributes to a sustainable energy future for India.



Ministry of Road Transport and Highways (MoRTH)

Develops and implements policies for the road transport sector, focusing on safety, efficiency, and infrastructure development. MoRTH plays a key role in promoting energy efficiency in transportation through initiatives like fuel efficiency standards for vehicles and promoting electric vehicle adoption



Ministry of Steel (MoS)

The ministry formulates national policies for the iron and steel sector in India. The MoS actively works to improve the energy efficiency of steel production processes and promotes the adoption of clean energy sources in steel manufacturing



Ministry of Mines (MoM)

The ministry regulates the mining sector in India, overseeing the exploration, development, and utilization of mineral resources critical to the energy sector, including coal, renewable energy materials, and rare earth elements. The MoM's role in integrated energy assessments is to ensure the sustainable management of these resources and inform policy decisions related to the energy transition



Ministry of Housing and Urban Affairs (MoHUA)

MoHUA formulates policies and programs related to housing and urban development in India. Their participation in integrated energy assessments is crucial for understanding energy consumption patterns in the buildings sector and developing strategies for sustainable urban development

Major policies and programs

This section highlights existing policies related to integrated energy assessments and their most prominent features. A more comprehensive list of policies is provided in the policy repository of the NITI Knowledge Platform.



Integrated Energy Policy, 2006:

The Integrated Energy Policy comprehensively covers all aspects of energy, encompassing energy security, accessibility, affordability, efficiency, and environmental concerns. The overarching goal of this energy policy is to ensure the safe and accessible provision of energy at the most competitive prices, with technical efficiency, economic viability, and environmental sustainability. This strategic vision aims to reliably meet the energy service demands of all sectors, including vulnerable households across the nation, in a sustainable and equitable manner.¹¹



National Mission for Enhanced Energy Efficiency, 2008:

The National Mission for Enhanced Energy Efficiency is a key national mission under the National Action Plan on Climate Change. It includes four major initiatives targeted at improving energy efficiency in energy-intensive industries: the Perform, Achieve, and Trade scheme for such industries, the Market Transformation for Energy Efficiency program focusing on efficient appliances and equipment, the Energy Efficiency Financing Platform aimed at facilitating investments in energy efficiency projects, and the Framework for Energy Efficient Economic Development to promote energy efficiency across different sectors. By implementing these initiatives, the NMEEE strives to boost energy

¹¹ International Energy Agency, Integrated Energy Policy, 2021

efficiency, curtail greenhouse gas emissions, and foster sustainable development in India.¹²



Green Credit Programme, 2023¹³:

Green Credit Programme, introduced by the Ministry of Environment, Forest and Climate Change, is an innovative market-based mechanism designed to incentivise voluntary environmental actions across diverse sectors. The Programme aims to generate green credits, which are tradable and available for trading on a domestic market platform. These green credits result from measures taken by a person or entity for the purposes of protection, preservation, or conservation of the environment such as tree plantation, water management, waste management, air pollution reduction, etc.



Energy Conservation (Amendment) Act, 2022

An amendment to the Energy Conservation Act in 2022 supports the formation of an Indian carbon market through an upcoming Carbon Credit Trading Scheme (CCTS). CCTS falls under the aegis of Bureau of Energy Efficiency (BEE), Ministry of Power, and Ministry of Environment, Forest & Climate Change. The objective is to decarbonise the Indian economy by pricing the Green House Gas (GHG) emission through trading of the Carbon Credit Certificates.¹⁴ Under this scheme, the BEE will identify sectors and assess the potential for reducing greenhouse gas emissions within those sectors. Additionally, it will establish targets for certain entities under a compliance mechanism.¹⁵



National Green Hydrogen Mission, 2023

The mission aims to position India as a global leader in the production, use and export of green hydrogen through (a) ramping up domestic capacity to annually produce green hydrogen up to a minimum of 5 MMT by 2030 by robust R&D efforts, reforming regulations, and pilot programs and (b) developing and scaling production-related technology, and build the infrastructure required for storage and delivery to make green hydrogen accessible and affordable.



Pradhan Mantri Kisan Urja Suraksha evam Utthaan Mahabhiyan (PM KUSUM), 2019⁶¹:

The PM KUSUM scheme aims to ensure energy security for farmers in India, along with fulfilling India's 2030 commitments of achieving 40% installed power capacity from non-fossil-fuel sources through the following key provisions:

- Component A provides for setting up 10 GW of grid-connected, decentralised RE power plants. These plants (500 kW to 2 MW capacity) can be set up by individual farmers, groups, and other specified entities, wherein DISCOMS would purchase generated power at pre-determined tariffs
- Component B provides for the installation of 17.5 lakh stand-alone solar agriculture pumps. Support will be provided to farmers for replacing existing diesel pumps with stand-alone solar water pumps (SWP) up to 7.5 HP in off-grid areas

¹² Ministry of Power, National Mission for Enhanced Energy Efficiency, 2021

¹³ Ministry of Environment, Forest and Climate Change, notification, 2023

¹⁴ Ministry of Power, Ministry of Power & Ministry of Environment, Forests & Climate Change to develop Carbon Credit Trading Scheme for Decarbonisation, 2023

¹⁵ Ministry of Power, Notification, 2023

- Component C provides for solarisation of 10 lakh grid-connected agriculture pumps. This will support individual farmers to solarise their pumps, and DISCOMS will purchase excess power at pre-fixed tariffs



Jawaharlal Nehru National Solar Mission, 2010¹⁰

The National Solar Mission aims to establish India as a global solar energy leader by (a) promoting the use of solar energy in various sectors through various schemes, favourable policies for solar deployment and incentives and (b) creating conducive conditions by scaling capacity and bringing about technological innovation to reduce costs.



Green Energy Corridor Scheme¹⁶

The Green Energy Corridor scheme aims to facilitate the evacuation and integration of renewable energy. It involves the developing transmission systems and control infrastructure to increase India's overall RE capacity by harnessing the potential of RE rich states like Andhra Pradesh, Gujarat, Himachal Pradesh, Jammu and Kashmir, Karnataka, Maharashtra, Rajasthan, Madhya Pradesh, and Tamil Nadu.



Amended) National Policy on Bio-Fuels, 2022¹⁷

The Policy enables the availability of biofuels in the market by:

- Proposing a target of 20% blending of ethanol in petrol Ethanol Supply Year (ESY) 2025-26
- Reducing the import of petroleum products by fostering domestic biofuel production



National Electricity Plan (NEP):

Published by the Ministry of Power, the NEP serves as a comprehensive blueprint for the Indian power sector's development. It outlines strategies for capacity addition, fuel security, grid expansion, and demand-side management. The NEP incorporates energy efficiency and renewable energy targets, fostering a more integrated approach to energy planning. By considering multiple energy sources and their interactions with the power grid, the NEP helps guide the development of a sustainable and reliable electricity system for India.¹⁸



India Climate and Energy Modeling Forum (ICEMF):

ICEMF serves as a collaborative platform bringing together researchers, policymakers, and stakeholders to develop and utilize energy-climate models. These models analyze the interactions between India's energy sector, economic growth, and climate change. ICEMF's work supports integrated energy assessments by providing valuable insights into the potential impacts of different energy policy choices on India's energy security, economic development, and environmental sustainability.

¹⁶ Ministry of Power, Transmission Works under Green Energy Corridors-I, Accessed December 2023

¹⁷ Ministry of Petroleum and Natural Gas, National Policy on Biofuels-2018 Amendment, 2022

¹⁸ Ministry of Power, National Electricity Plan, 2023



National Action Plan on Climate Change (NAPCC):

Launched in 2008, the NAPCC is a key national strategy to address climate change. It comprises eight national missions, including the National Mission for Enhanced Energy Efficiency (NMEEE) and the National Mission for Sustainable Habitat (NMSH). These missions promote energy-efficient technologies, green buildings, and sustainable urban development, all contributing to a more integrated approach to energy management and climate action in India.

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Way forward

This section highlights considerations for policymakers seeking to advance policy formulation and implementation in this sector. These are based on reports and materials prepared by key government bodies and expert institutions like the NITI Aayog, CEEW, TERI, and IEEFA.

 Strengthen energy policy coordination to ensure energy security	- Establish deeper energy policy coordination across different ministries and agencies through increased use of high-level inter-ministerial mechanisms that align and oversee energy policies. - Foster collaboration between the central and state governments to align their energy policies through frequent dialogues and consultations, while also leveraging existing bilateral and multilateral partnerships (such as the US-India Strategic Clean Energy Partnership) to exchange best practices, improve technical assistance delivery, and drive innovation in energy policymaking.
 Promote renewable energy to aid India's energy transition	- Advance the use of integrated assessment models (which simulate dynamic energy systems and their interaction with other systems like land, water, climate, and the economy) to effectively evaluate trade-offs across renewable energy options and pathways. - Encourage investment in renewable energy projects through favourable policies and incentives for RE deployment, and especially for emerging technologies such as offshore wind, floating solar, green hydrogen, and energy storage. Such measures – spanning tax benefits, low-interest and concessional debt, and standardised power purchase agreements for RE producers – should be aligned and coordinated with broader state and national development goals. - Enhance grid infrastructure to accommodate the increasing share of variable renewable energy and enhance its flexibility and stability. This can be done by strengthening the transmission and distribution network, implementing the Green Energy Corridor project, and deploying smart grid solutions (inclusive of demand response, load management, and grid balancing efforts).
 Improve energy access to foster socio-economic development	- Expand the grid infrastructure to remote and rural areas based on updated and highly disaggregated energy access assessments that identify barriers and drivers and energy access and use. These can be conducted by national and state bodies through a variety of survey, census, satellite imagery, and geospatial analysis tools. - Promote context-specific and user-centric energy access and use solutions. For example, policy support can ensure that decentralised renewable energy solutions like solar home systems and mini-grids evaluate reliability, affordability, quality, and safety, in policy design, and include the voice of local stakeholders like communities, civil society organizations, and entrepreneurs in their delivery. - Evaluate and compare the impacts and co-benefits of different energy access options and scenarios. Energy access options and scenarios can be

	evaluated and compared, using appropriate indicators and methods, to assess their impacts and co-benefits on other systems and sectors, such as economy, environment, health, education, and gender. National and state policymakers can draw on the research and analysis of domestic and international think tanks to evaluate and compare the impacts and co-benefits of different energy access options and scenarios.
Invest in capacity building for energy policymakers to drive effective energy governance	- Develop training and education programs for policymakers on IEA concepts and methods, such as systems thinking, scenario analysis, multi-criteria analysis, and cross-sector stakeholder engagement. IEA training and education can also expose energy policymakers to the global best practices and experiences and facilitate peer-to-peer learning and exchange. - Strengthen the data and information systems for IEA by enhancing the quality, availability, and usability of data on the energy system and its interactions with other systems, and leverage technology to create robust and real-time energy data management systems. Expand the capacity of central and state planning bodies to conduct advanced forecasting and modelling for informed decision-making and policy analysis. - Create a network of IEA experts and practitioners across states to provide technical assistance, guidance, and advice to the energy policymakers, and support them in addressing the challenges and opportunities of IEA. A network of IEA experts and practitioners can also foster collaboration and partnership among IEA experts and practitioners from different sectors, disciplines, and backgrounds, such as academia, research, industry, civil society, and media, to generate and disseminate knowledge and innovation for IEA. India can benefit from the existing networks and initiatives, such as the India Energy Modelling Forum, the India Smart Grid Forum, and the India Energy Storage Alliance, to create such networks.

Recommended readings

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